CSC301 – Practice Midterm

Closed book, closed notes, no electronics.

Question (5pts): Name one of the programming language paradigms/classes discussed in class.

Question (5 pts): What are the defining characteristics/hallmarks of the programming language paradigm/class you mentioned in the previous question?

Question (5pts): Given the following types P and Q as

P = {"Pancake", "Dragon", "Whisper", "Guitar", "Sunshine", "Elephant", "Moonlight"}
Q = {“Pancake”, “Moonlight”, “Elephant”}

Is there a subtype-supertype relationship between P and Q? Why? Why not?

Question (5 pts): Assume that B < A, that is, B is a subtype of A. Formally determine with type inferencing whether the following program is a type safe program,

```
Declare x and y of type B
Declare z of type A
Let z = (x | y)
```

Given two values the '|' operator randomly picks one of them and returns it.

Question (10 pts) **:** Consider the following Asteroid function,

```
function funny
    with [] do
        []
    with [x|y] do
        [x] + funny(y)
end
```

What does this function do?

Question 2 (10pts – 5pts each): Consider the following Asteroid function,

```
load system math.

function g
   with (x,y):(%real,%integer) do
      math@pow(x,y) -- raise x to the power of y
   with _ do
      throw Error("unknown value")
end
```

(a) What would be the result of the call g(3.0, 2)?

(b) What would be the result of the call g(1.0, 2, 3)?